

DSA 8401 — Applied Machine Learning

Exam Preparation Question Bank

Multiple-Choice Questions — Topics 1 to 9

108 questions · 12 per topic · four options each, one correct answer

Instructions

- This paper covers the nine core applied-machine-learning topics of the course, from end-to-end pipeline fundamentals through to modern Generative AI (RAG).
- Each topic contains 12 questions. Distractors are written to be plausible — read every option carefully, including numbers and named methods, before choosing.
- Work through each section, then check your answers against the Answer Key with explanations at the end of this document. Do not consult the key until you have committed to an answer for every question in a section.
- Content is drawn from the course syllabus, lecture notes (Foundations, Data Preparation, Evaluation, Ensembles, Unsupervised Learning, Deep Learning, CNNs) and the nine practical lab notebooks (End-to-End ML, Messy Data, Evaluation, Trees/Optuna, Unsupervised, MLP, CNN X-ray, RNN COVID-19 forecasting, RAG).

Topic Overview

Topic 1. Foundations & End-to-End ML Pipelines — *ML lifecycle, bias-variance trade-off, no-free-lunch, production system failure modes, batch vs online learning*

Topic 2. Advanced Data Preparation & Feature Engineering — *Missingness mechanisms, leakage-safe pipelines, target encoding, entity resolution, cyclical encoding, power transforms*

Topic 3. Classical ML Evaluation & Imbalanced Classes — *Accuracy paradox, precision/recall/F1/ROC-AUC/PR-AUC, threshold selection under a cost matrix, calibration, stratified CV*

Topic 4. Ensemble Methods & Hyperparameter Tuning — *Bagging vs boosting, out-of-bag evaluation, random forests, gradient boosting sensitivity, Optuna/TPE, feature importance*

Topic 5. Unsupervised Learning & Dimensionality Reduction — *K-Means, DBSCAN, silhouette score, PCA and explained variance, t-SNE/UMAP limitations, clusters-are-not-labels*

Topic 6. Deep Learning Fundamentals (MLP) — *Architecture, activation functions, backpropagation, loss functions, Adam, regularisation, class imbalance*

Topic 7. Convolutional Neural Networks & Computer Vision — *Convolution mechanics, padding/stride, pooling, augmentation caveats, transfer learning with MobileNetV2, evaluation priorities*

Topic 8. Sequence Models: RNNs, LSTMs & Time-Series Forecasting — *Series preparation, windowing, leakage-safe time-series splits, SimpleRNN vs LSTM/GRU, baseline comparison*

Topic 9. Modern GenAI: Transformers, LLMs & Retrieval-Augmented Generation — *The RAG problem/pipeline, chunking, cosine similarity, retrieval vs generation failure, evaluation, temperature*

Topic 1

Foundations & End-to-End ML Pipelines*ML lifecycle, bias-variance trade-off, no-free-lunch, production system failure modes, batch vs online learning*

1.1 A model shows high training error and high validation error, and the two are close together. What is the correct diagnosis and the correct remedy?

- A. High variance (overfitting); the remedy is to add more data
- B. High bias (underfitting); the remedy is a richer feature set or a more flexible model family
- C. High variance (overfitting); the remedy is to increase regularization strength further
- D. Irreducible error; nothing can be done about it

1.2 According to the no-free-lunch theorem as it is used in this course, what is the correct practical implication?

- A. Since no algorithm is universally best, hyperparameter tuning is pointless
- B. Deep learning will eventually outperform every other model family given enough data
- C. Because no model family wins a priori, several diverse families should be shortlisted and compared using honest validation rather than one being anointed in advance
- D. Gradient-boosted trees are mathematically proven optimal for all tabular data

1.3 In production ML systems, the CACE principle ('Changing Anything Changes Everything') refers to which phenomenon?

- A. The requirement to containerize a model before it can be deployed
- B. The fact that features interact through the model's learned weights, so improving one input feature can silently degrade how the model uses every other feature
- C. The need to retrain models continuously in an online-learning setup
- D. The certification checklist required before a model reaches production

1.4 A credit-scoring model declines applicants scoring below a cutoff. Regarding hidden feedback loops, which statement is most accurate?

- A. Feedback loops only occur in online-learning systems, never in batch systems
- B. Declined applicants' subsequent repayment behaviour is never observed, so the model's belief about that segment can never be corrected by future training data
- C. Feedback loops are automatically eliminated by k-fold cross-validation
- D. Feedback loops only affect regression models, not classifiers

1.5 Per the most-cited production-ML systems paper referenced in this course, the ML code itself in a mature deployment is typically:

- A. The majority of the total codebase
- B. A small fraction of the total system, with the rest being data collection, verification, feature extraction, serving infrastructure, monitoring, configuration, and resource management
- C. Irrelevant to overall system reliability
- D. The only component that requires version control

1.6 Streaming fraud detection on payment rails is best matched to which learning paradigm in the batch-vs-online trade-off, and why?

- A. Batch learning, because it offers stronger reproducibility and audit trails

- B. Online (incremental) learning, because it tracks fast drift and validates prequentially (test-then-train) around every update
- C. Neither; fraud detection cannot be automated
- D. Batch learning, because online learning cannot process streaming data

1.7 In the end-to-end ML lifecycle taught in this course, hyperparameters should be tuned using:

- A. The test set, since it gives the most honest estimate
- B. The training set only, with no validation step needed
- C. Cross-validation folds carved out of the training data, while the sealed test set is reserved for a single final evaluation
- D. A randomly reshuffled mix of train and test data at every epoch

1.8 Which of the following is a hyperparameter rather than a model parameter learned during training?

- A. The coefficients of a fitted logistic regression
- B. The regularization strength $C (=1/\lambda)$ in logistic regression
- C. The weights of a trained neural network
- D. The split thresholds of a fitted decision tree

1.9 In logistic regression, increasing the regularization strength (larger λ , i.e. smaller C) typically has which effect?

- A. It decreases both training loss and test loss monotonically
- B. It increases training loss but can lower expected (test) risk by reducing variance — this is the bias-variance trade-off in action
- C. It has no effect on bias or variance
- D. It always causes underfitting regardless of the magnitude chosen

1.10 In the ML lifecycle taught in this course, the 'fine-tune and select' step refers to:

- A. Cleaning missing values before any modelling begins
- B. Hyperparameter search and cross-validated comparison among shortlisted model families, prior to a single sealed test-set evaluation
- C. Deploying the chosen model to a FastAPI endpoint
- D. Collecting additional raw data from source systems

1.11 A team observes a suspiciously perfect validation score immediately after adding two new engineered features. What is the appropriate first reaction, per this course's treatment of model trust?

- A. Celebrate — the two features are clearly powerful predictors
- B. Ship the model immediately, since perfect validation implies perfect production performance
- C. Treat the improvement with suspicion and investigate for data leakage before trusting it
- D. Increase the regularization strength to push the score even higher

1.12 Which option correctly reflects the 'small green box' view of a deployed ML system?

- A. Feature extraction is part of monitoring infrastructure, not the ML system
- B. Configuration and resource management are irrelevant to reproducing a deployed model
- C. Data collection, verification, feature extraction, serving infrastructure, monitoring, configuration, and resource management all surround a comparatively small ML-code core
- D. The ML code represents the majority of a mature deployment's components

Topic 2

Advanced Data Preparation & Feature Engineering

Missingness mechanisms, leakage-safe pipelines, target encoding, entity resolution, cyclical encoding, power transforms

2.1 In the mobile-money lab, `gps_lat` and `gps_lon` are missing precisely when a customer has disabled location sharing, and that behaviour is itself correlated with fraud risk. This missingness is best classified as:

- A. MCAR (missing completely at random)
- B. MAR (missing at random, conditional on other observed variables)
- C. MNAR (missing not at random) — missingness depends on the unobserved value or behaviour itself
- D. Not a missing-data problem, since the column technically exists

2.2 Which is the correct leakage-safe procedure for scaling numeric features in a train/test workflow?

- A. Fit the scaler on the full dataset (train+test combined) before splitting, for consistency
- B. Fit the scaler on the training split only, then transform the validation and test splits with it
- C. Fit a separate, independent scaler on the training, validation, and test sets
- D. Scaling never needs to be 'fit'; recompute it fresh for every batch

2.3 Why is naive (non-out-of-fold) target encoding dangerous?

- A. It cannot handle high-cardinality categorical variables
- B. Each row's own target value leaks into the very statistic used to encode that row (self-leakage); the remedy is out-of-fold encoding, where each row is encoded using statistics that exclude it
- C. It only works on numeric features, never categorical ones
- D. It requires a GPU to compute

2.4 In the mobile-money fraud lab, `manual_review_score` and `settlement_status` must be dropped before training because:

- A. They are highly correlated with `is_fraud` and therefore trivially predictive
- B. They are only written after a transaction has already been scored and investigated, so they would not be available at prediction (serving) time
- C. They contain missing values that cannot be imputed
- D. They are string columns requiring additional encoding work

2.5 Which best captures the formal definition of data leakage used in this course?

- A. Any missing value present in the training data
- B. A validation accuracy above 90%
- C. The use, during training or model selection, of information (a statistic or feature) that would not genuinely be available at serving/prediction time
- D. Leaking personally identifiable information to unauthorized parties

2.6 In the mobile-money fraud lab, what happened to ROC-AUC when the two leaky columns were added back into the feature set?

- A. It dropped from roughly 0.63 to roughly 0.50
- B. It stayed at approximately 0.63, since the honest features were already the most predictive
- C. It jumped from approximately 0.63 to approximately 1.00 — about +0.37 of pure illusion
- D. It became undefined, since ROC-AUC cannot exceed 0.90

2.7 Why must customers be resolved to a canonical customer_id via reg_id (the KYC national registration key) rather than via msisdn (mobile number)?

- A. msisdn is never recorded in the dataset
- B. A single customer may hold several mobile numbers, so grouping by msisdn would incorrectly split one customer into several entities
- C. reg_id is easier to hash computationally than msisdn
- D. msisdn formats cannot be parsed across countries

2.8 Why does the pipeline evaluate with GroupKFold grouped by customer_id rather than plain KFold?

- A. GroupKFold trains noticeably faster than KFold
- B. It ensures no customer appears in both a training fold and a validation fold, preventing customer-level information from leaking across the split
- C. GroupKFold automatically balances class imbalance
- D. It removes the need for a held-out test set entirely

2.9 Why encode hour-of-day using sine and cosine transforms instead of the raw integer 0-23?

- A. To reduce the total number of columns needed
- B. Because the raw integer implies hour 23 and hour 0 are numerically far apart, when in reality they are adjacent in time; the sine/cosine pair preserves that cyclical adjacency
- C. Because scikit-learn models cannot accept integer-valued features
- D. To convert a numeric feature into a categorical one

2.10 The raw amount column contains values like '1,500/-', '(227/-)', and '4,742/- Dr'. What must the parsing logic correctly handle regarding sign?

- A. Every value should be treated as positive regardless of formatting
- B. Parenthesised values and values with a 'Dr' (debit) suffix represent negative/debit amounts and must be converted to negative numbers, unlike the plain credit format
- C. Only the comma thousand-separators need to be stripped; sign is irrelevant
- D. All three formats are equivalent and can be parsed identically without special-casing

2.11 The PowerTransformer with method='yeo-johnson' is preferred over Box-Cox in a feature-engineering pipeline that may include which type of values?

- A. Only strictly positive values
- B. Zero or negative values — Yeo-Johnson, unlike Box-Cox, is defined for zero and negative inputs as well as positive ones
- C. Only categorical values
- D. Only values that are already normally distributed

2.12 RFM (recency, frequency, monetary) customer-level features are computed 'as of a fixed SCORING_TS timestamp.' What is the primary purpose of anchoring to a fixed scoring timestamp?

- A. To make the feature-engineering code run faster
- B. To ensure every feature only uses information that would genuinely have been available at the moment of scoring, preventing temporal leakage from future transactions
- C. To standardise timezones across countries
- D. To remove the need for any train/test split

Topic 3

Classical ML Evaluation & Imbalanced Classes

Accuracy paradox, precision/recall/F1/ROC-AUC/PR-AUC, threshold selection under a cost matrix, calibration, stratified CV

3.1 On the loan dataset roughly 6% of borrowers default. A model that predicts 'no default' for every single applicant scores 94% accuracy. What does this illustrate?

- A. The model is excellent, since 94% is a high score
- B. Accuracy is a poor metric under class imbalance because the majority class dominates the score, masking a completely useless model
- C. Accuracy and recall are always identical for imbalanced data
- D. A 94%-accurate model must also have a high ROC-AUC

3.2 Of the applicants a model flags as likely defaulters, the fraction that actually default is:

- A. Recall
- B. Precision
- C. Specificity
- D. F1 score

3.3 Why is PR-AUC generally preferred over ROC-AUC when the positive class (e.g. defaulters) is rare?

- A. PR-AUC is computationally cheaper
- B. ROC-AUC can look misleadingly flattering when true negatives are abundant, since it doesn't emphasise how rare the positive class is; PR-AUC focuses squarely on the positive class's precision-recall trade-off
- C. ROC-AUC cannot be computed at all on imbalanced datasets
- D. PR-AUC and ROC-AUC always produce identical model rankings

3.4 Under a cost matrix where a missed default costs roughly ten times what a wrongly rejected good applicant costs, the lab found the optimal decision threshold lands near:

- A. 0.50 (the scikit-learn default)
- B. 0.90
- C. 0.23
- D. 0.05

3.5 A reliability diagram plots predicted probability against observed frequency. A well-calibrated model's curve should:

- A. Be a flat horizontal line at 0.5
- B. Lie on the diagonal — predicted probability approximately equals observed frequency
- C. Always lie strictly above the diagonal
- D. Be undefined for any probabilistic classifier

3.6 What does wrapping a classifier in CalibratedClassifierCV with isotonic regression check and correct for?

- A. Whether the model's feature importances are stable across seeds
- B. Whether the predicted probabilities actually reflect true outcome frequencies, adjusting them if the raw scores are over- or under-confident
- C. Whether the training data contains duplicate rows
- D. Whether the model is overfitting the training set

3.7 Why does the lab use StratifiedKFold with five folds instead of plain KFold on the loan default data?

- A. StratifiedKFold trains roughly five times faster
- B. It preserves the ~6% positive class rate within every fold, so no fold is left with too few (or zero) positive examples
- C. StratifiedKFold removes the need to report a standard deviation across folds
- D. It automatically applies SMOTE oversampling within each fold

3.8 Per this course's methodology, cross-validated results should be reported as:

- A. A single number taken from whichever fold performed best
- B. Mean plus or minus one standard deviation across folds, never a single number from a single split
- C. Only the maximum score observed across all folds
- D. The median of the training-set scores only

3.9 A classifier achieves precision = 0.40 and recall = 0.80 on a held-out fold. What is its F1 score, to two decimal places?

- A. 0.60
- B. 0.53
- C. 0.50
- D. 0.44

3.10 An ROC-AUC of 0.5 on the loan default problem means:

- A. The model perfectly separates defaulters from non-defaulters
- B. The model ranks a randomly chosen defaulter above a randomly chosen non-defaulter no better than a coin flip — i.e. random guessing
- C. The model is exactly 50% accurate
- D. The model has 50% precision

3.11 Which of the two metrics, ROC-AUC or F1, is described as insensitive to the choice of decision threshold?

- A. F1, because it is a harmonic mean
- B. ROC-AUC, because it summarises ranking quality across all possible thresholds rather than performance at one fixed cutoff
- C. Both are equally threshold-dependent
- D. Neither metric involves a threshold at any point

3.12 A lender's business team insists on keeping the classification threshold at the scikit-learn default of 0.5 'because that's the standard.' What is the strongest course-grounded critique of this stance?

- A. 0.5 is mathematically proven optimal for all binary classifiers
- B. The 0.5 threshold is an arbitrary convention, not a law; the optimal operating point should be chosen by sweeping thresholds against an explicit, business-specific cost matrix
- C. Thresholds should never be changed once a model has been trained
- D. 0.5 only applies to regression models, so the critique does not apply here

Topic 4

Ensemble Methods & Hyperparameter Tuning

Bagging vs boosting, out-of-bag evaluation, random forests, gradient boosting sensitivity, Optuna/TPE, feature importance

4.1 Which statement correctly distinguishes bagging from boosting?

- A. Bagging trains base learners sequentially; boosting trains them in parallel
- B. Bagging trains base learners in parallel on bootstrap samples to reduce variance; boosting trains them sequentially, each correcting the previous one's errors, to reduce bias
- C. Bagging and boosting are two names for the same algorithm
- D. Bagging primarily reduces bias; boosting primarily reduces variance

4.2 A bootstrap sample of size n is drawn with replacement from a training set of size n . Approximately what fraction of rows are left out of a given bootstrap sample (out-of-bag)?

- A. About 10%
- B. About 25%
- C. About 36.8% (approximately $1/e$)
- D. About 50%

4.3 The course notes that 'OOB is not out-of-time.' What key limitation does this flag?

- A. OOB error can only be computed after training completes, wasting computation
- B. The OOB estimate randomises over rows, not over time, so it can miss temporal degradation that a genuinely deployed model would experience
- C. OOB scores are always lower than cross-validated scores
- D. OOB estimation requires a GPU

4.4 An unpruned single decision tree trained to purity on the training set typically exhibits:

- A. High bias and low variance, generalising very well
- B. Low bias and high variance — it memorises the training set, including its noise, and generalises poorly
- C. Neither bias nor variance, since trees are non-parametric
- D. Guaranteed better test performance than a random forest

4.5 Beyond bootstrap sampling of rows, what additional randomisation does a random forest introduce at each split, which a plain bagged ensemble of trees does not?

- A. Random shuffling of the target labels
- B. A random subset of features is considered at each split, further decorrelating the individual trees
- C. Random removal of the deepest tree layers
- D. Random reweighting of the loss function per tree

4.6 Compared to random forests, gradient boosting (and XGBoost/LightGBM) on tabular data tends to:

- A. Perform worse in nearly all cases and is rarely used in practice
- B. Often outperform random forests, but is far more sensitive to hyperparameters, risking overfitting if the learning rate is too high or too many estimators are used
- C. Be completely insensitive to the learning rate hyperparameter
- D. Require no hyperparameter tuning at all

4.7 Why is Optuna's Tree-structured Parzen Estimator (TPE) sampler generally more sample-efficient than grid search or random search?

- A. TPE exhaustively evaluates every possible hyperparameter combination
- B. TPE models the distribution of good versus bad configurations from past trials and samples new candidates where improvement is likely, unlike grid search (which wastes trials on unpromising regions) or random search (which doesn't learn from past trials)
- C. TPE requires no objective function to be defined
- D. TPE always converges in exactly 50 trials regardless of the search space

4.8 Which combination correctly describes the lab's Optuna study setup?

- A. 50 trials, a MedianPruner for early stopping of unpromising trials, objective = mean cross-validated PR-AUC, direction = maximize
- B. 500 trials, no pruner, objective = accuracy, direction = minimize
- C. 5 trials, a GridPruner, objective = training loss, direction = maximize
- D. 50 trials, objective = ROC-AUC only, direction = minimize

4.9 Which feature-importance method is described as biased toward high-cardinality features, and which is described as measured on held-out data and therefore more trustworthy?

- A. Built-in Gini importance is biased toward high-cardinality features; permutation importance (measured on held-out data) is more trustworthy
- B. Permutation importance is biased toward high-cardinality features; Gini importance is more trustworthy
- C. SHAP values are biased toward high-cardinality features; Gini importance is unbiased
- D. Both are equally unbiased and fully interchangeable

4.10 What distinguishes SHAP values from the other two feature-importance views (Gini and permutation) discussed in the ensembles lab?

- A. SHAP only works for random forests, not gradient boosting
- B. SHAP gives per-prediction attributions that also show the direction of an effect, rather than a single global ranking
- C. SHAP cannot be computed for tree-based models
- D. SHAP is mathematically identical to permutation importance

4.11 Which pair of hyperparameters in the lab's Optuna search space specifically controls L1 and L2 regularisation respectively?

- A. subsample and colsample_bytree
- B. reg_alpha (L1) and reg_lambda (L2)
- C. max_depth and min_child_weight
- D. n_estimators and learning_rate

4.12 A student tunes an XGBoost model and finds that increasing n_estimators from 200 to 2000 while keeping a relatively high learning_rate of 0.3 causes validation PR-AUC to fall even though training PR-AUC keeps rising. What is the most likely explanation?

- A. This is impossible; boosting cannot overfit
- B. The model is overfitting — boosting is highly sensitive to the number of estimators combined with a high learning rate, and this combination pushes the ensemble to fit noise in the training residuals
- C. The scoring metric (PR-AUC) is broken and should be replaced with accuracy
- D. This indicates underfitting, so even more estimators are needed

Topic 5

Unsupervised Learning & Dimensionality Reduction

*K-Means, DBSCAN, silhouette score, PCA and explained variance, t-SNE/UMAP limitations, clusters-are-not-labels***5.1 Each iteration of Lloyd's K-Means algorithm alternates between which two steps?**

- A. Compute the silhouette score; prune outliers
- B. Assign each point to its nearest centroid; then move each centroid to the mean of its newly assigned points
- C. Fit a Gaussian to each cluster; then compute eigenvalues
- D. Randomly reinitialise centroids; then merge overlapping clusters

5.2 Which of the following is NOT a documented assumption or sensitivity of K-Means in this course?

- A. Sensitive to feature scale, requiring standardisation first
- B. Sensitive to centroid initialisation, addressed with k-means++ and multiple restarts (n_init)
- C. Assumes clusters are roughly spherical and of similar size
- D. Automatically detects the correct number of clusters k without any input

5.3 The silhouette score ranges from -1 to 1 and measures:

- A. The inertia (within-cluster sum of squares) alone
- B. How close each point is to its own cluster compared with the nearest other cluster; higher values indicate better-defined clusters
- C. The number of noise points identified
- D. The explained variance ratio of the clustering

5.4 On the mobile-money customer-segmentation dataset, the silhouette score peaks at k equal to:

- A. 2
- B. 3
- C. 4
- D. 8

5.5 Which capability does DBSCAN have that K-Means lacks?

- A. DBSCAN requires the number of clusters k to be specified in advance, just like K-Means
- B. DBSCAN can find clusters of arbitrary shape, does not require k in advance, and explicitly labels outliers as noise (label -1), making it a natural fit for anomaly detection
- C. DBSCAN always produces exactly the same clusters as K-Means on identical data
- D. DBSCAN cannot handle datasets containing outliers

5.6 The two key parameters of DBSCAN are:

- A. k (number of clusters) and n_init
- B. eps (neighbourhood radius) and min_samples
- C. learning_rate and n_estimators
- D. perplexity and n_components

5.7 In the unsupervised learning lab, how many of the original 24 features were needed to explain 95% of the variance via PCA?

- A. 24
- B. 18
- C. 11

D. 3

5.8 Consistent with the leakage-safe practice taught throughout this course, PCA in a modelling pipeline should be:

- A. Fit on the combined train+test data for maximum stability
- B. Fit on the training set only, then applied (transformed) onto the test set
- C. Refit separately and independently on train and test
- D. Fit on the test set to ensure the components generalise

5.9 Which statement about t-SNE and UMAP, as taught in this course, is correct?

- A. They are linear methods suitable as a preprocessing step before a downstream classifier
- B. Their axes are not interpretable, distances between well-separated clusters are not meaningful, and they should be used for visualisation only — never as preprocessing feeding a classifier
- C. They always outperform PCA for feature extraction in supervised pipelines
- D. They require the number of output dimensions to match the number of input features

5.10 The unsupervised lab's 'headline caution' about a cluster showing a high observed fraud rate is that such a cluster is:

- A. Definitive proof the clustering algorithm has learned to detect fraud
- B. A hypothesis to investigate, not a fraud detector — clusters are not labels
- C. Evidence the K-Means clustering should replace the supervised fraud classifier entirely
- D. Irrelevant and should be discarded without further analysis

5.11 If total variance is the trace of the covariance matrix S (the sum of all eigenvalues), the explained variance ratio of the first d principal components is defined as:

- A. The largest eigenvalue divided by the number of features
- B. The sum of the first d eigenvalues divided by the sum (trace) of all eigenvalues
- C. The number of components d divided by the total number of features
- D. The product of the first d eigenvalues

5.12 A colleague argues: 'The K-Means cluster with the highest fraud rate can be used directly as our production fraud classifier.' What is the strongest course-grounded objection?

- A. K-Means clusters are supervised labels and can indeed be used directly for classification
- B. Clustering is unsupervised and reflects density/distance patterns in feature space, not ground-truth fraud; a high-fraud cluster is a lead for investigation, and a proper supervised evaluation against the withheld label is still required before any deployment decision
- C. K-Means clusters cannot be evaluated against the fraud label at all, even for a sanity check
- D. There is no objection — this is exactly how the lab recommends deploying the model

Topic 6

Deep Learning Fundamentals (MLP)

*Architecture, activation functions, backpropagation, loss functions, Adam, regularisation, class imbalance***6.1 Lab 06's MLP architecture for the tabular loan classification problem is:**

- A. Dense(128) -> Dense(64) -> Dense(1)
- B. Dense(64, relu) -> Dropout(0.3) -> Dense(32, relu) -> Dropout(0.3) -> Dense(1, sigmoid)
- C. Conv2D(32) -> MaxPooling2D -> Dense(1)
- D. LSTM(64) -> Dense(1, softmax)

6.2 ReLU is preferred over sigmoid/tanh in hidden layers primarily because:

- A. ReLU outputs are always between 0 and 1, matching probability requirements
- B. ReLU's gradient is exactly 1 for positive inputs, so it does not saturate and does not squash the gradient signal flowing backward through many layers, avoiding the vanishing-gradient problem sigmoid suffers from
- C. ReLU has no known failure modes
- D. ReLU is smooth and differentiable everywhere, unlike sigmoid

6.3 The 'dying ReLU' failure mode occurs when:

- A. A unit's input is always negative, so it outputs zero forever and its weights stop updating; Leaky ReLU is the usual remedy
- B. The learning rate is set to exactly zero
- C. Too few training epochs are used
- D. The sigmoid function saturates at the output layer

6.4 Which loss function and output activation are correctly matched to Lab 06's binary classification setup?

- A. Categorical cross-entropy with a softmax output
- B. Binary cross-entropy with a sigmoid output
- C. Mean squared error with a ReLU output
- D. Hinge loss with a linear output

6.5 What makes the Adam optimiser 'much less fussy to tune than plain SGD', per the course material?

- A. Adam always uses a fixed learning rate of 0.1 regardless of the problem
- B. Adam adapts a per-parameter learning rate using running estimates of the first and second moments of the gradient
- C. Adam does not require backpropagation
- D. Adam eliminates the need for a loss function

6.6 Lab 06 combines three regularisation mechanisms together. Which of the following is NOT one of them?

- A. Dropout at 0.3
- B. L2 weight decay
- C. Early stopping with patience=10 and restore_best_weights=True
- D. Batch normalisation after every dense layer

6.7 Per the course's framing, why is early stopping described as 'the cheapest form of regularisation available'?

- A. It requires purchasing additional compute resources

- B. It costs nothing computationally beyond monitoring validation loss and requires no tuning beyond the patience value
- C. It is the only regularisation method that works for tabular data
- D. It replaces the need for a validation set entirely

6.8 Passing `class_weight` to `model.fit()` in Keras to up-weight the minority class is described as the neural-network equivalent of which XGBoost parameter?

- A. `n_estimators`
- B. `scale_pos_weight`
- C. `max_depth`
- D. `reg_lambda`

6.9 Per the course's 'honest conclusion' comparing the MLP to the tuned gradient-boosted model on tabular loan data, which statement is correct?

- A. Neural networks always beat gradient boosting on tabular data
- B. On tabular data, gradient boosting is usually still the right default; neural networks earn their keep on unstructured data such as images, audio, text, and sequences
- C. Gradient boosting cannot handle class imbalance, unlike MLPs
- D. The MLP trains faster and is easier to interpret than gradient boosting

6.10 The shift-invariance property states $\text{softmax}(z + c \cdot \mathbf{1}) = \text{softmax}(z)$ for any constant c . What is the practical significance of this property?

- A. It means softmax outputs can be negative
- B. It allows implementations to subtract the maximum logit from all logits before exponentiating, for numerical stability, without changing the output probabilities
- C. It proves softmax is mathematically identical to the sigmoid function
- D. It means the loss function no longer needs true labels

6.11 Binary cross-entropy loss is mathematically equivalent to:

- A. The mean squared error between predicted and true labels
- B. The negative log-likelihood of the true label under a Bernoulli distribution parameterised by the predicted probability
- C. The Euclidean distance between the logits and the labels
- D. The silhouette score of the predicted classes

6.12 A student trains an MLP and observes training loss steadily decreasing while validation loss decreases initially, plateaus, then starts rising after epoch 15, even though training continues to epoch 100. Which combination from Lab 06's regularisation toolkit most directly addresses this?

- A. Increase the learning rate and remove dropout entirely
- B. Early stopping monitoring validation loss (restoring the best weights around epoch 15) combined with dropout and/or L2 weight decay to reduce overfitting
- C. Switch the loss function to mean squared error
- D. This pattern indicates underfitting and requires a larger model with no regularisation

Topic 7

Convolutional Neural Networks & Computer Vision

Convolution mechanics, padding/stride, pooling, augmentation caveats, transfer learning with MobileNetV2, evaluation priorities

7.1 A 224x224 greyscale image flattened into a Dense layer of 128 units needs over six million weights in the first layer alone. Convolutional layers avoid this explosion through:

- A. Reducing the image resolution before the first layer
- B. Parameter sharing — sliding a small learned kernel across the image and reusing the same weights at every spatial position
- C. Removing the colour channels entirely
- D. Using a larger batch size

7.2 Translation invariance, a property convolution provides, means:

- A. The network's output never changes no matter how the image is cropped
- B. A feature is detected regardless of where in the frame it appears
- C. The image is automatically rotated into a canonical orientation before processing
- D. Every pixel is treated as equally important regardless of the convolutional receptive field

7.3 Lab 07's CNN uses three convolutional blocks, each Conv2D(filters, 3x3, relu) -> MaxPooling2D(2x2). The filter counts across the three blocks are:

- A. 16, 16, 16 (constant)
- B. 128, 64, 32 (decreasing)
- C. 32, 64, 128 — doubling at each block, while pooling halves the spatial dimensions
- D. 8, 256, 8

7.4 Using the standard convolution output-size formula for input size n , kernel size f , padding p , and stride s : for a 224x224 input, a 3x3 kernel, 'same' padding ($p=1$), and stride 1, the output spatial size is:

- A. 224x224 — unchanged, because 'same' padding with stride 1 preserves spatial dimensions
- B. 222x222
- C. 112x112
- D. 226x226

7.5 Taking padding $p=0$ ('valid' padding) with kernel size f at unit stride causes the output feature map to:

- A. Stay exactly the same size as the input
- B. Shrink by $f-1$ relative to the input
- C. Double in size
- D. Become undefined

7.6 In Lab 07's chest X-ray classification, horizontal flips are deliberately NOT used as a data augmentation, because:

- A. Horizontal flips are computationally too expensive
- B. A chest X-ray has a definite left and right (the heart sits on one side), so flipping would create anatomically impossible training images
- C. Horizontal flips only work for colour images, not greyscale
- D. Keras does not support horizontal flipping

7.7 Lab 07's transfer-learning approach with MobileNetV2 involves:

- A. Training the entire network from scratch on the X-ray dataset with random initial weights
- B. Loading MobileNetV2 pretrained on ImageNet with `include_top=False`, freezing the convolutional base, and training only a new classification head; fine-tuning later unfreezes the top block at a reduced learning rate
- C. Using MobileNetV2 purely as a feature-visualisation tool with no training at all
- D. Replacing MobileNetV2's convolutional layers with fully connected layers before training

7.8 Despite ImageNet containing no medical images, transfer learning from it to chest X-rays works well because:

- A. ImageNet secretly contains medical images
- B. Low-level edge and texture filters learned on ImageNet transfer well to new visual domains, even one as different as medical imaging, which is especially valuable on a very small target dataset
- C. MobileNetV2 was specifically designed for medical imaging
- D. Transfer learning only works when source and target datasets are visually identical

7.9 In the head 'GlobalAveragePooling2D -> Dense(64, relu) -> Dropout(0.5) -> Dense(1, sigmoid)', what does GlobalAveragePooling2D primarily accomplish compared to a Flatten layer?

- A. It increases the number of parameters flowing into the dense head
- B. It collapses each feature map to a single average value, drastically reducing parameters compared to flattening and helping reduce overfitting
- C. It applies max pooling instead of average pooling
- D. It removes the need for any convolutional layers

7.10 Why is a confusion matrix reported alongside accuracy in the chest X-ray lab, with particular attention to false negatives?

- A. False negatives are computationally cheaper to compute
- B. In a medical screening context, a missed pneumonia case (false negative) is far more costly than a false positive, so the confusion matrix's breakdown matters more than a single accuracy number
- C. Accuracy is always sufficient for medical applications
- D. False positives are always more dangerous than false negatives in medical screening

7.11 A ResNet-50 backbone is described as having a total stride of 32 at its deepest features. What does 'total stride of 32' mean?

- A. There are exactly 32 convolutional layers in ResNet-50
- B. Each spatial position in the deepest feature map corresponds to a 32x32 region of the original input image, since the spatial resolution has been cumulatively downsampled by a factor of 32
- C. The network processes 32 images per batch
- D. The kernel size used throughout ResNet-50 is 32x32

7.12 A student fine-tunes MobileNetV2 but sets the fine-tuning learning rate to the same value used for training the randomly-initialised head (e.g. 1e-3) instead of a much smaller value (e.g. 1e-5). What is the most likely consequence?

- A. No consequence; the learning rate for fine-tuning pretrained layers does not matter
- B. The large gradient updates at a high learning rate risk catastrophically destroying the pretrained convolutional weights' useful low-level features, effectively undoing the benefit of transfer learning
- C. This will make training converge faster with no downside
- D. This only affects the classification head, never the pretrained base

Topic 8

Sequence Models: RNNs, LSTMs & Time-Series Forecasting

Series preparation, windowing, leakage-safe time-series splits, SimpleRNN vs LSTM/GRU, baseline comparison

8.1 The COVID-19 lab converts a cumulative confirmed-case column into new daily cases through which operation?

- A. Multiplying each day's count by the previous day's count
- B. Taking the day-to-day difference (first-order differencing) of the cumulative series
- C. Dividing each day's cumulative count by 7
- D. Taking the logarithm of the cumulative count

8.2 After differencing, the lab (a) clips negative values to zero and (b) smooths with a 7-day rolling average. What do these two steps respectively address?

- A. (a) removes duplicate rows; (b) removes missing values
- B. (a) corrects for downward-revised cumulative counts that would otherwise produce impossible negative 'new cases'; (b) smooths uneven day-of-week reporting patterns (e.g. fewer cases reported at weekends)
- C. (a) removes outlier countries; (b) converts the series into a classification target
- D. Both steps are purely cosmetic and do not affect modelling

8.3 With LOOKBACK = 30, how is a single supervised training example (X, y) constructed from the daily case series?

- A. X is a single day's value; y is the average of the next 30 days
- B. X is the previous 30 days of values; y is the very next day's value
- C. X is the entire series; y is a randomly chosen day from it
- D. X and y are both 30-day windows offset by 30 days

8.4 Why must the train/test split for the COVID-19 forecasting lab be done by date rather than randomly?

- A. Random splits are computationally more expensive
- B. A random split would let the model train on future values and predict the past, which is a form of leakage; splitting by date (last 60 days held out) keeps the evaluation honest
- C. scikit-learn does not support random time-series splits
- D. Random splitting would violate data protection regulations

8.5 Consistent with the course's leakage-safe practice, the MinMaxScaler used in Lab 08 should be fit on:

- A. The full series, including the test period, for the most accurate scaling
- B. The training portion only, then applied (transformed) onto the test portion
- C. The test portion only
- D. Refit fresh for every single prediction

8.6 The core weakness of a SimpleRNN over long sequences is:

- A. It cannot process sequences of variable length
- B. Vanishing gradients — the gradient signal shrinks multiplicatively as it is backpropagated through time, preventing the network from learning dependencies more than a few steps back
- C. It has too many parameters and always overfits
- D. It cannot be trained using backpropagation at all

8.7 An LSTM's cell state is regulated by which three gates?

- A. Input, hidden, output
- B. Forget, input, output
- C. Reset, update, candidate
- D. Encoder, decoder, attention

8.8 The LSTM's cell state gives the gradient what kind of path through time, and why does this matter?

- A. A multiplicative path, identical to SimpleRNN, so there is no real difference
- B. An additive path through time (rather than a purely multiplicative one), which is why the LSTM can learn long-range dependencies that a SimpleRNN struggles with
- C. No path at all; the LSTM does not use backpropagation
- D. A path that only exists during inference, not training

8.9 A GRU is described in the course material as:

- A. A heavier variant of the LSTM with four gates and two cell states
- B. A lighter variant with two gates and no separate cell state
- C. Mathematically identical to a SimpleRNN
- D. Usable only for image data, not sequences

8.10 Why does the lab compare SimpleRNN and LSTM forecasts against a naive 'tomorrow equals today' baseline?

- A. The naive baseline is expected to always perform worst and serves no real purpose
- B. This baseline is surprisingly hard to beat on a smoothed series, and any forecasting model that fails to beat it has demonstrated nothing of value
- C. The naive baseline is only used to test the code before deep learning models are built
- D. The naive baseline replaces the need for MAE as an evaluation metric

8.11 The reported evaluation metric for the COVID-19 forecasting lab is MAE (mean absolute error) in cases per day, rather than, say, ROC-AUC. Why is this the appropriate choice?

- A. ROC-AUC is always superior for any machine learning task
- B. This is a regression/forecasting problem (predicting a continuous count), not a classification problem, so an error metric like MAE in the target's own units is appropriate rather than a classification ranking metric
- C. MAE can only be used with RNNs, never with classical models
- D. MAE and ROC-AUC are mathematically identical for time-series data

8.12 A student builds a SimpleRNN to forecast case counts and finds it performs about as well as the naive baseline, while an LSTM trained on the identical windowed data beats the baseline by a meaningful margin. What is the most likely underlying reason for this gap?

- A. The LSTM simply has more parameters, and more parameters always guarantee better performance regardless of the task
- B. The 30-day lookback window likely requires modelling dependencies further back in the sequence than the SimpleRNN's vanishing-gradient-limited effective memory can reach, whereas the LSTM's gated cell state preserves that longer-range signal
- C. SimpleRNNs cannot be applied to numeric time series at all, so the comparison is invalid
- D. The naive baseline is a form of data leakage that only affects the SimpleRNN

Topic 9

Modern GenAI: Transformers, LLMs & Retrieval-Augmented Generation

The RAG problem/pipeline, chunking, cosine similarity, retrieval vs generation failure, evaluation, temperature

9.1 What core limitation of large language models does Retrieval-Augmented Generation primarily address?

- A. LLMs cannot generate grammatically correct text
- B. LLMs only know what was in their training data, so when asked about material such as course notes or post-cutoff events, they will often produce a fluent, confident and wrong answer — a hallucination
- C. LLMs cannot process any numeric data
- D. LLMs require a GPU to run at inference time

9.2 Compared to fine-tuning, why does the course describe RAG as 'the cheaper answer'?

- A. Fine-tuning is free while RAG always costs money
- B. Fine-tuning is expensive and slow, needs retraining whenever documents change, and still does not let the model cite a source, whereas RAG leaves the model's weights alone and fetches relevant text at question time
- C. RAG requires no embedding model at all, unlike fine-tuning
- D. Fine-tuning and RAG solve entirely unrelated problems

9.3 Which correctly orders the offline indexing phase of a RAG system?

- A. Embed the question -> retrieve chunks -> load documents -> chunk them
- B. Load the documents -> chunk them into passages -> embed each chunk into a vector -> store the vectors
- C. Generate an answer -> chunk the answer -> store the answer as a vector
- D. Store the vectors -> load the documents -> chunk them -> embed the question

9.4 In the online query phase, after the user's question is embedded with the same embedding model used for indexing, what happens next?

- A. The LLM is fine-tuned on the question immediately
- B. The top-k chunks are retrieved by similarity to the question vector, a prompt is built containing those chunks plus the question, and the LLM generates an answer instructed to use only the supplied context
- C. The question is discarded and a random chunk is returned
- D. The documents are re-chunked using a different chunk size for every query

9.5 What is the central trade-off in choosing chunk size for a RAG system?

- A. Larger chunks are always strictly better, with no downside
- B. Chunks too small get retrieved without enough surrounding context to answer from; chunks too large dilute the embedding (one vector representing several unrelated ideas) and waste prompt tokens
- C. Chunk size has no effect on retrieval quality, only on storage cost
- D. Chunk size must always equal exactly one sentence

9.6 Why does RAG retrieval compare embedding vectors using cosine similarity rather than Euclidean distance?

- A. Cosine similarity is always faster to compute than any other distance metric
- B. Direction carries the semantic meaning of an embedding, not magnitude; if vectors are L2-normalised to unit length, cosine similarity reduces to a simple dot product, making retrieval over the whole index one matrix multiplication
- C. Euclidean distance cannot be computed for high-dimensional vectors

D. Cosine similarity is only defined for binary vectors

9.7 What problem does adding a modest overlap between consecutive chunks solve?

- A. It reduces the total number of chunks needed
- B. It stops a fact that spans a chunk boundary from being split across two chunks and effectively lost to retrieval
- C. It removes the need for an embedding model
- D. It guarantees the LLM will never hallucinate

9.8 RAG's two failure modes should be diagnosed separately. Which correctly distinguishes them?

- A. Retrieval failure is when the model hallucinates numbers; generation failure is when the embedding model crashes
- B. Retrieval failure is when the right chunk was never fetched (measured by hit rate/recall at k); generation failure is when the right chunk was fetched but the model still answered badly (measured by faithfulness)
- C. Both failure modes are measured by the exact same metric, so no distinction is needed
- D. Retrieval failure only happens with GPT-family models, never with open-source LLMs

9.9 Why does the course emphasise diagnosing which of the two RAG failure modes you have before changing anything?

- A. Because fixing a retrieval failure by editing the prompt is wasted effort — the underlying problem (the right chunk never reached the model) is untouched by prompt changes
- B. Because prompt engineering always fixes both failure modes simultaneously
- C. Because retrieval failures cannot be measured at all
- D. Because generation failures are impossible once retrieval succeeds

9.10 Why is a temperature of 0 typically the right sampling choice for a RAG system, per the course glossary?

- A. Temperature 0 makes the embedding model run faster
- B. Temperature 0 makes output near-deterministic, which is desirable when the answer should be pinned to the retrieved context rather than creative
- C. Temperature only affects image-generation models, not text
- D. Temperature 0 disables retrieval entirely

9.11 Lab 09 deliberately builds RAG using no framework such as LangChain or LlamaIndex, and no vector database. What is the stated pedagogical reason?

- A. Frameworks are prohibited in academic coursework
- B. So that every moving part of the pipeline (embedding, storage, similarity search, prompt construction) stays visible rather than hidden behind framework abstractions
- C. LangChain and LlamaIndex are incompatible with numpy
- D. Vector databases cannot store text embeddings

9.12 A team's RAG chatbot for course-notes Q&A is evaluated with hit-rate@5 = 0.95 (the correct chunk is almost always retrieved) but faithfulness = 0.40 (many answers contain unsupported claims). Following the course's diagnostic principle, what should the team prioritise fixing?

- A. Increase k and re-embed the documents with a different embedding model, since retrieval must be the problem
- B. The problem is primarily a generation failure, not a retrieval failure — since the right chunks are already being found, the team should focus on the generation step, e.g. strengthening the instruction to use only supplied context, lowering temperature toward 0, and checking that the correct chunks are actually included in the constructed prompt
- C. Rebuild the entire vector index from scratch, since a hit-rate@5 of 0.95 indicates a broken embedding model

D. Faithfulness cannot be improved once hit-rate is high, so the issue is unsolvable

Answer Key & Explanations

Check your work topic by topic. Each explanation states why the correct option is right.

Topic 1 — Foundations & End-to-End ML Pipelines

1.1 Correct answer: B

High error on both sets that are close together is the signature of underfitting (high bias): the model family is too rigid to represent the true function, so more flexibility or richer features are needed, not more regularization or more data.

1.2 Correct answer: C

No-free-lunch is read constructively in this course: priors (inductive biases) are unavoidable, so shortlist diverse families and let cross-validated comparison decide empirically — 'gradient boosting usually wins on tabular data' is an empirical regularity, never a theorem.

1.3 Correct answer: B

CACE names entanglement: because features interact through learned weights, no feature or config change is ever truly local — improving one input can silently degrade the model's use of the others.

1.4 Correct answer: B

This is the classic hidden feedback loop: the model's own decisions (declining an applicant) shape what future training data can even exist, since a declined segment's true repayment behaviour is never observed again.

1.5 Correct answer: B

The 'small green box' diagram shows the ML code as a comparatively tiny core surrounded by a much larger system of data collection, verification, feature extraction, serving infrastructure, monitoring, configuration, and resource management.

1.6 Correct answer: B

The course's batch-vs-online table pairs 'streaming fraud on payment rails' with online learning specifically because it tracks fast-moving drift, at the cost of harder reproducibility/audit and a live-model attack surface.

1.7 Correct answer: C

Hyperparameters are tuned on CV folds; the test set stays sealed until final evaluation — using the test set for tuning is itself a form of leakage (test-set overfitting in disguise).

1.8 Correct answer: B

C (equivalently λ) is set before training and controls how training proceeds — it is a hyperparameter, not a value learned by gradient descent or a tree-growing algorithm.

1.9 Correct answer: B

Stronger regularization trades a higher training loss for lower variance; whether expected risk falls depends on where on the bias-variance curve the model currently sits — it is a trade-off, not a guaranteed improvement at every magnitude.

1.10 Correct answer: B

'Fine-tune and select' is the model-selection step: search hyperparameters and compare shortlisted families via cross-validation, before the sealed test set is finally used once.

1.11 Correct answer: C

A suspiciously perfect score is treated as evidence of leakage, not of skill — this mirrors the ROC-AUC jumping from ~0.63 to ~1.00 when leaky columns were reintroduced in the fraud lab.

1.12 Correct answer: C

The diagram's point is precisely that the ML code is the small green box; nearly everything else in a mature deployment is supporting infrastructure — 'a great notebook is not a system.'

Topic 2 — Advanced Data Preparation & Feature Engineering

2.1 Correct answer: C

Because the propensity to disable location sharing (and hence to have missing GPS) is itself tied to the customer's fraud-relevant behaviour, this is MNAR — the mechanism the lab explicitly assigns to `gps_lat/gps_lon`.

2.2 Correct answer: B

The rule is 'split, then fit, then apply everywhere': fitting a scaler on the full dataset before splitting is explicitly named as a leakage cause in the course material.

2.3 Correct answer: B

Target encoding is described as 'the most powerful classical encoder and also the most dangerous' precisely because of this self-leakage; out-of-fold (or a shrinkage prior) is the neutralising fix.

2.4 Correct answer: B

Both fields are post-outcome: they get written after a human review that only happens once a transaction is already flagged, so a live scoring model could never see them at the moment it needs to make a prediction.

2.5 Correct answer: C

Formally, leakage exists when the training-time feature map differs from — has more information than — the feature map genuinely available at serving time; this mismatch inflates offline scores that then collapse in production.

2.6 Correct answer: C

This is the lab's headline result: ROC-AUC climbs from ~0.63 (honest features) to ~1.00 once the two post-outcome leaky columns are reintroduced — a suspiciously perfect score signalling leakage, not skill.

2.7 Correct answer: B

Entity resolution matters because one customer, one `reg_id`, can have several `msisdn` numbers; using `msisdn` as the grouping key would fragment a single customer's history across multiple pseudo-entities.

2.8 Correct answer: B

`GroupKFold` keeps every row belonging to one `customer_id` entirely inside one fold, so the model can never be validated on a customer it has already implicitly seen during training — mirroring the leakage-prevention theme of the whole lab.

2.9 Correct answer: B

Cyclical (sine/cosine) encoding fixes the false discontinuity a raw hour integer creates at the midnight wraparound, letting distance-based and linear models see hour 23 and hour 0 as close together.

2.10 Correct answer: B

This is standard accounting notation: parentheses and a 'Dr' suffix both denote a debit (negative), while the plain '1,500/-' form denotes a credit (positive) — missing this distinction silently corrupts the amount feature.

2.11 Correct answer: B

Box-Cox requires strictly positive inputs; Yeo-Johnson generalises the transform to handle zero and negative values too, which matters for features like signed transaction amounts.

2.12 Correct answer: B

Anchoring feature computation to a fixed `SCORING_TS` is the temporal analogue of the general leakage-safe rule: each customer's RFM stats must reflect only what was knowable up to that point in time, not future transactions.

Topic 3 — Classical ML Evaluation & Imbalanced Classes

3.1 Correct answer: B

This is the textbook accuracy paradox: a trivial always-predict-majority model scores 94% while catching zero defaulters, which is why accuracy is abandoned in favour of precision/recall/ROC-AUC/PR-AUC on this dataset.

3.2 Correct answer: B

Precision is defined over predicted positives: of everything flagged, what fraction is truly positive. Recall instead asks, of all true positives, what fraction was caught.

3.3 Correct answer: B

With rare positives, ROC-AUC can stay high largely because true negatives are easy to rank correctly, understating how poorly the model handles the minority class — PR-AUC is the metric the course prefers in this setting.

3.4 Correct answer: C

Sweeping the threshold against an explicit 10:1 cost matrix (missed default vs. wrongly rejected good applicant) moved the optimal operating point down to roughly 0.23, far from the arbitrary 0.5 default.

3.5 Correct answer: B

Calibration means predicted probabilities mean what they claim: when the model says 0.7, roughly 70% of such cases should actually be positive, which traces the diagonal on a reliability diagram.

3.6 Correct answer: B

Calibration is orthogonal to discrimination: a model can rank cases well (good ROC-AUC/PR-AUC) yet have poorly calibrated probability outputs; isotonic-regression calibration remaps raw scores onto the diagonal of the reliability diagram.

3.7 Correct answer: B

With only ~6% positives, a plain random KFold risks folds with very few defaulters; stratification keeps the class ratio consistent across all five folds so each fold gives a meaningful estimate.

3.8 Correct answer: B

The course insists on reporting mean $\pm$ one standard deviation across folds, because a single split's number hides the variability that cross-validation exists to reveal.

3.9 Correct answer: B

$F1 = 2 \cdot P \cdot R / (P + R) = 2(0.40)(0.80) / (0.40 + 0.80) = 0.64 / 1.20 \approx 0.53$. Note that the arithmetic mean of 0.40 and 0.80 would incorrectly give 0.60 — F1 is the harmonic mean, which is pulled toward the smaller value.

3.10 Correct answer: B

ROC-AUC is the probability a random positive is ranked above a random negative; 0.5 is the value achieved by random guessing, and 1.0 is perfect ranking — it says nothing directly about accuracy or precision.

3.11 Correct answer: B

ROC-AUC integrates performance across every possible threshold, so it doesn't depend on picking one operating point; F1 (like precision and recall) is computed at a specific chosen threshold and changes if that threshold moves.

3.12 Correct answer: B

The lab explicitly frames 0.5 as an arbitrary convention and derives a business-appropriate threshold (≈ 0.23) from a real cost matrix — the critique should target exactly that gap between convention and cost-informed choice.

Topic 4 — Ensemble Methods & Hyperparameter Tuning

4.1 Correct answer: B

This is the core distinction taught in the ensembles module: bagging = parallel bootstraps → variance reduction (e.g. random forest); boosting = sequential residual-correction → bias reduction (e.g. gradient boosting).

4.2 Correct answer: C

As n grows, the probability a given row is never drawn in n draws-with-replacement approaches $(1 - 1/n)^n \rightarrow 1/e \approx 0.368$, leaving roughly 36.8% of rows out-of-bag for each tree.

4.3 Correct answer: B

OOB gives a free, row-wise honest estimate, but because it does not respect time ordering, it cannot substitute for a time-aware validation scheme when temporal drift is a real production risk.

4.4 Correct answer: B

A fully grown, unpruned tree can perfectly fit training data (low bias) but is highly sensitive to the particular training sample (high variance), which is exactly what motivates averaging many such trees in a random forest.

4.5 Correct answer: B

Random forests add feature-subsampling at every split on top of row-bootstrapping; this extra randomisation decorrelates trees more than bagging alone, improving the variance-reduction benefit of averaging.

4.6 Correct answer: B

Boosting 'usually wins on tabular data' but is fragile: a learning rate that's too high paired with too many estimators drives the sequential residual-fitting process to fit noise, unlike the more forgiving averaging in random forests.

4.7 Correct answer: B

TPE is a Bayesian-style sequential search: it builds probabilistic models of promising vs. unpromising regions from prior trials, focusing subsequent sampling where the objective is likely to improve — a genuine efficiency gain over grid/random search.

4.8 Correct answer: A

The lab runs 50 trials with a MedianPruner that halts unpromising trials early; the objective returns mean CV PR-AUC and the study direction is 'maximize', matching the imbalanced-classification focus carried over from the evaluation module.

4.9 Correct answer: A

The lab explicitly contrasts three views: Gini importance (biased toward high-cardinality splits, computed only on training data), permutation importance (computed on held-out data, hence more trustworthy), and SHAP (per-prediction, directional).

4.10 Correct answer: B

Unlike a single global importance ranking, SHAP decomposes each individual prediction into per-feature contributions, including their sign/direction — a strictly more granular view than Gini or permutation importance.

4.11 Correct answer: B

`reg_alpha` applies L1 (sparsity-inducing) regularisation and `reg_lambda` applies L2 (shrinkage) regularisation to the boosting objective — both are explicitly listed among the searched hyperparameters.

4.12 Correct answer: B

This is exactly the fragility the course warns about: boosting will overfit if the learning rate is too high or there are too many estimators, producing a widening training/validation gap — the classic overfitting signature.

Topic 5 — Unsupervised Learning & Dimensionality Reduction

5.1 Correct answer: B

K-Means is the classic assign-then-update loop: assign points to the nearest centroid, recompute each centroid as the mean of its assigned points, and repeat until convergence, minimising within-cluster sum of squares (inertia).

5.2 Correct answer: D

K-Means requires k to be specified in advance — choosing it correctly is a separate task (elbow method, silhouette score); K-Means itself has no built-in mechanism to discover k on its own.

5.3 Correct answer: B

Silhouette compares intra-cluster cohesion against nearest-cluster separation for each point, on a -1-to-1 scale, giving a per-point (and averaged) measure of how well-separated the clustering is — distinct from inertia, which silhouette is used alongside.

5.4 Correct answer: C

The lab reports that the silhouette score peaks at $k = 4$ on this dataset, which the lab uses alongside the elbow method to settle on the number of segments.

5.5 Correct answer: B

DBSCAN groups by density rather than distance-to-centroid, so it naturally handles non-spherical clusters, needs no pre-specified k , and treats sparse regions as noise (-1) rather than forcing every point into a cluster.

5.6 Correct answer: B

DBSCAN is parameterised by ϵ , the radius defining a point's neighbourhood, and min_samples , the minimum number of neighbours required for a point to be a core point — no k is involved.

5.7 Correct answer: C

The lab reports that 11 of the original 24 components are sufficient to reach the 95%-explained-variance threshold, a substantial dimensionality reduction while retaining almost all the informative variance.

5.8 Correct answer: B

PCA follows the same split-then-fit-then-apply discipline as scalers and other preprocessing: fitting on the full dataset before splitting would leak test-set variance structure into the training-time transformation.

5.9 Correct answer: B

t-SNE and UMAP are explicitly flagged as non-linear, visualisation-only tools: their distances (especially between separated clusters) do not carry quantitative meaning, so feeding their output into a classifier is discouraged, unlike PCA.

5.10 Correct answer: B

The lab's closing caution is explicit: cross-tabulating cluster assignment against the withheld `is_fraud` flag is a sanity check, not a validation of a fraud detector — an unsupervised clustering is not trained against any label.

5.11 Correct answer: B

Because $\text{tr}(S)$ equals the sum of all eigenvalues (total variance) and each principal component's variance equals its corresponding eigenvalue, the explained variance ratio for the top d components is the ratio of their eigenvalue sum to the trace.

5.12 Correct answer: B

This directly restates the lab's headline caution: a cluster is a hypothesis, not a detector; deploying it as a classifier skips the supervised evaluation (precision/recall/ROC-AUC against ground truth) that any production fraud model requires.

Topic 6 — Deep Learning Fundamentals (MLP)

6.1 Correct answer: B

The lab's exact architecture is two hidden Dense-ReLU blocks each followed by 0.3 dropout, ending in a single sigmoid output unit for binary classification.

6.2 Correct answer: B

Sigmoid saturates at both tails (gradient $\rightarrow 0$), which is the vanishing-gradient problem; ReLU's constant gradient of 1 on the positive side keeps gradient signal flowing through deep stacks of layers.

6.3 Correct answer: A

A ReLU unit whose pre-activation is always negative always outputs zero and has zero gradient, so it can never recover via gradient descent — the course names Leaky ReLU as the standard fix.

6.4 Correct answer: B

Binary classification with a single output unit uses a sigmoid activation (to produce a probability in $[0,1]$) paired with `binary_crossentropy` loss, exactly as specified for the loan-default MLP.

6.5 Correct answer: B

Adam maintains per-parameter moving averages of the gradient (first moment) and squared gradient (second moment) to adaptively scale each parameter's step size, which is why it tends to work well with less manual learning-rate tuning than vanilla SGD.

6.6 Correct answer: D

The three mechanisms explicitly named are dropout, L2 weight decay, and early stopping; batch normalisation is not part of Lab 06's stated regularisation stack for this MLP.

6.7 Correct answer: B

Early stopping just watches validation loss during training that's already happening and halts/reverts when it stops improving — essentially free, unlike dropout rates or weight-decay coefficients, which need their own tuning.

6.8 Correct answer: B

Both mechanisms address class imbalance the same way — by inflating the loss contribution of minority-class examples — so `class_weight` in Keras is drawn as the direct analogue of `scale_pos_weight` in XGBoost.

6.9 Correct answer: B

The MLP reaches comparable PR-AUC to the tuned gradient-boosted baseline but takes longer to train and is harder to interpret — reinforcing the course's broader claim that boosting remains the default for tabular problems.

6.10 Correct answer: B

Shift invariance is exactly what licenses the standard 'max-subtraction trick' used to prevent overflow when exponentiating large logits — the resulting probabilities are unchanged.

6.11 Correct answer: B

Binary cross-entropy is derived directly from maximum-likelihood estimation for a Bernoulli-distributed target: minimising cross-entropy is equivalent to maximising the Bernoulli log-likelihood of the observed labels.

6.12 Correct answer: B

The described curve is the textbook overfitting signature (diverging train/validation loss); early stopping directly counters it by reverting to the best validation-loss epoch, while dropout/L2 reduce the model's tendency to overfit in the first place.

Topic 7 — Convolutional Neural Networks & Computer Vision

7.1 Correct answer: B

A convolutional kernel (e.g. 3x3) is applied identically at every location, so the number of learned weights depends on the kernel size and channel depth, not the image's spatial dimensions — a dramatic parameter reduction versus a fully-connected layer.

7.2 Correct answer: B

Because the same kernel/weights slide across the whole image, a learned pattern (e.g. an edge) is detected wherever it occurs spatially, rather than requiring separate weights per location.

7.3 Correct answer: C

The architecture follows the standard CNN pattern: filter count doubles (32→64→128) as spatial resolution is halved by pooling at each block, trading spatial extent for representational depth.

7.4 Correct answer: A

'Same' padding is defined precisely to keep output size equal to input size at unit stride: $\text{output} = \text{floor}((n + 2p - f)/s) + 1 = \text{floor}((224 + 2 - 3)/1) + 1 = 224$.

7.5 Correct answer: B

With $p=0$ and $s=1$, $\text{output} = n - f + 1$, i.e. the map shrinks by exactly $f-1$ pixels per dimension — this is what 'valid' padding means in Keras.

7.6 Correct answer: B

This is a domain-specific augmentation caveat: unlike natural photographs, a chest X-ray's left-right asymmetry (heart position) is medically meaningful, so a flip would train the model on physiologically impossible images.

7.7 Correct answer: B

The lab's two-stage recipe is standard transfer learning: freeze the pretrained base and train a new head first, then unfreeze the top block for a gentle fine-tune at a much lower learning rate ($1e-5$).

7.8 Correct answer: B

The general-purpose low-level filters (edges, textures, gradients) learned from natural images turn out to be broadly useful visual primitives, letting a small medical dataset benefit hugely from features learned on an unrelated large dataset.

7.9 Correct answer: B

GlobalAveragePooling2D reduces each channel's entire spatial map to one number (its average), so the dense head that follows has far fewer input connections — and hence far fewer parameters to overfit with — than a Flatten layer would produce.

7.10 Correct answer: B

The clinical cost asymmetry (a missed diagnosis is worse than a false alarm) means the model's false-negative rate deserves explicit scrutiny, which a single aggregate accuracy figure would hide.

7.11 Correct answer: B

Total stride accumulates the downsampling of every stride-2 operation through the network; a total stride of 32 means the deepest feature map is $1/32$ the spatial resolution of the input, with each unit's receptive field spanning roughly a 32×32 input region.

7.12 Correct answer: B

The whole point of using a much smaller learning rate ($1e-5$) during fine-tuning is to nudge the pretrained weights gently; a head-sized learning rate applied to unfrozen pretrained layers can wipe out the transferred features almost immediately.

Topic 8 — Sequence Models: RNNs, LSTMs & Time-Series Forecasting

8.1 Correct answer: B

Differencing a cumulative series (today minus yesterday) is the standard way to recover the daily increment from a running total — exactly the transformation applied before windowing.

8.2 Correct answer: B

Real-world reporting artefacts (backward corrections producing negative diffs, and weekday/weekend reporting rhythms) are cleaned up by clipping and smoothing respectively, before the series is turned into supervised training windows.

8.3 Correct answer: B

A sliding window of the previous LOOKBACK=30 days forms the input X , and the target y is the single following day's value — this windowing step is what converts a raw time series into a supervised learning problem.

8.4 Correct answer: B

Time series has an inherent temporal ordering; a random split would mix future information into the training set, letting the model implicitly 'see the future' — hence the strict date-based split (last 60 days as test).

8.5 Correct answer: B

Scaling follows the same fit-on-train-only discipline used throughout the course (for scalars, PCA, target encoders, etc.) — fitting on the full series including the test window would leak test-period statistics into training.

8.6 Correct answer: B

SimpleRNN backpropagation-through-time repeatedly multiplies gradients by the same recurrent weight matrix, causing the signal to shrink (or explode) exponentially with sequence length — this is what limits its effective memory.

8.7 Correct answer: B

LSTM cells use a forget gate (what to discard), an input gate (what to write), and an output gate (what to read out) to control the cell state — the reset/update gates belong to a GRU, not an LSTM.

8.8 Correct answer: B

The gated cell state lets gradient flow additively (through the forget-gate-controlled update) rather than through repeated matrix multiplications, which is precisely why LSTMs mitigate the vanishing-gradient problem SimpleRNNs suffer from.

8.9 Correct answer: B

The GRU simplifies the LSTM by merging the cell state into the hidden state and using only two gates (reset and update), trading some expressiveness for fewer parameters and faster training.

8.10 Correct answer: B

A smoothed, slowly-changing series makes 'predict yesterday's value' a deceptively strong baseline; the lab insists any sequence model must clear this bar to demonstrate genuine forecasting skill rather than trivially tracking recent history.

8.11 Correct answer: B

ROC-AUC evaluates ranking quality for classification; forecasting a continuous daily case count is a regression task, for which an error metric like MAE, expressed in the target's natural units (cases/day), is the meaningful choice.

8.12 Correct answer: B

This connects the lab's two central facts directly: SimpleRNN's vanishing gradients limit its effective memory to a handful of steps, while the LSTM's additive cell-state path lets it exploit the full 30-day lookback — explaining the observed performance gap.

Topic 9 — Modern GenAI: Transformers, LLMs & Retrieval-Augmented Generation

9.1 Correct answer: B

RAG exists precisely because an LLM's knowledge is frozen at training time; asked about material outside that (private documents, recent events), it tends to hallucinate a fluent but wrong answer rather than admit ignorance.

9.2 Correct answer: B

The course contrasts the two directly: fine-tuning bakes knowledge into weights (costly, stale as soon as documents change, no citations), while RAG retrieves fresh text per query and can point to its source.

9.3 Correct answer: B

Indexing is a one-time offline pipeline: load documents, chunk into passages, embed each chunk, and store the resulting vectors — done once, before any user query arrives.

9.4 Correct answer: B

The query-time pipeline is: embed the question with the same embedder → retrieve top-k similar chunks → assemble a context-plus-question prompt → generate an answer grounded in that retrieved context.

9.5 Correct answer: B

Chunking sits on a trade-off curve: too-small chunks lose context needed to actually answer the question, while too-large chunks blur multiple ideas into a single embedding vector and bloat the prompt — a few hundred words with modest overlap is the usual starting point.

9.6 Correct answer: B

Embeddings encode meaning in the direction of the vector; cosine similarity captures that directly, and with unit-normalised vectors it collapses to a dot product — enabling retrieval as one efficient matrix multiply across the whole index.

9.7 Correct answer: B

Without overlap, a fact sitting right at a chunk boundary can be split so that neither chunk alone contains it fully; a modest overlap keeps such boundary-straddling facts intact within at least one chunk.

9.8 Correct answer: B

The course draws a firm line between the retrieval stage (did the pipeline fetch the right passage — hit rate@k) and the generation stage (given the right passage, did the model answer faithfully) — each needs its own metric and its own fix.

9.9 Correct answer: A

If the correct chunk was never retrieved, no amount of prompt tweaking can produce a faithful answer, since the model never had the information; correctly diagnosing the failure mode first avoids wasting effort on the wrong fix.

9.10 Correct answer: B

RAG wants the model to faithfully report what the retrieved context says rather than generate novel or varied phrasing; a temperature near 0 minimises sampling randomness, keeping the output tightly grounded in the supplied passages.

9.11 Correct answer: B

Building RAG from raw numpy and HTTP calls is an explicit pedagogical choice: it forces every step of the pipeline — chunking, embedding, cosine-similarity search, prompt assembly — to be implemented and understood directly, not hidden behind a framework's abstractions.

9.12 Correct answer: B

High hit-rate@5 already shows retrieval is working well; low faithfulness with good retrieval is the signature of a generation failure, so effort belongs on the generation step (prompting, temperature, verifying the prompt actually contains the retrieved context) rather than on retrieval infrastructure.